GinCol Lab · Aula

Student worksheet

Biology, Geology and Environmental Sciences · 1st year of Bachillerato · Inside the Earth · Class 4 of 5 · With Nil

What was in the shadow

Class video: https://youtu.be/vM3mxG9WK5M

The code leads to the video.

Name: _____ Date: _____

Before watching

1. If you have ever looked from the side at a spoon or a pencil standing in a glass of water, what happened to it? Why do you think that happens?

2. In the previous class, between what distances from the earthquake was the shadow of the P wave? What arrived inside the shadow?

3. **Your prediction.** A wave arrives diagonally at a rock where it travels more slowly. Does it carry on straight, bend towards the perpendicular to the boundary, or bend away from it? Draw it.

4. **Your prediction.** In the second class you saw that the core is, on average, twice as dense as the Earth. In a denser material, do you think waves travel faster or more slowly? Why?

While you watch the video

1. How far did the direct P wave reach? What are the two puzzles this class solves?

2. Why does a wave bend when it passes into a rock where it travels more slowly? Explain it with the row of people arm in arm.

3. Read Snell's law: what happens to the angle with the perpendicular if the second rock is slower? And if it is faster? And if the speeds are the same? And if the wave arrives head-on?

4. Why does the spoon in the glass look broken? What has bent?

5. What did Mohorovičić see in 1909? How can a wave that takes a detour arrive first?

6. At what depth did he put the boundary? What is it called?

7. Why do the rays curve in the mantle? If the whole Earth were mantle, would there be a shadow?

8. **Your bet.** What happens to the P rays that enter the core? A: they cross it without bending. B: they all bounce off. C: they bend towards the centre. Why?

9. If the rays cross the core, why is there a shadow? How is the core like a magnifying glass?

10. What did Oldham propose in 1906? What did Gutenberg calculate? What fraction of the Earth's volume does the core take up, and why?

11. What happens to S waves when they reach the core? Why is their shadow much bigger than that of the P?

12. **Stop and think.** If no S wave crosses the core, what does that tell us about it?

13. What did Jeffreys look at in 1926? Why is it evidence independent of the evidence from S waves?

14. Is it true that the denser a rock is, the faster waves travel? What happens at the edge of the core, and why?

15. Which earthquake did Inge Lehmann study? What reached the Soviet stations inside the shadow?

16. Why did she make a model with straight rays? How were those waves explained before?

17. **Your hypothesis.** What would you put inside the core to make some rays land in the shadow?

18. What was Lehmann's hypothesis? Why do those rays land in the shadow? What does the title of her paper, "P′", mean?

19. Why is the curve over the Fiji data a prediction and not a fit?

20. Why has the first wave that reaches Toledo from a big New Zealand earthquake crossed the inner core? How long does it take?

21. Did Lehmann say that the inner core is solid? Who proposed it, and how was it confirmed?

22. What was expected of her? What did she choose? How was she treated?

23. **Stop and think.** What do you think it takes for someone to be able to use all their strengths?

24. How can we know something that no one will ever see? What prediction was risked, and did it come true?

In class: Lehmann's model, with ruler, compass and calculator

In 1936, Inge Lehmann did the sums for a very simple model of the Earth by hand. You are going to redo them and check your numbers against the ones she printed in her paper, "P′".

The model. A mantle where the P wave travels at 10 km/s and a core of 5/9 of the Earth's radius where it travels at 8 km/s. The rays are straight and only bend at the edge of the core, following Snell's law. The focus of the earthquake is at the surface.

The angles, with the names used in her paper:

- i_0: the angle at which the ray leaves the focus, measured from the vertical;

- i_1: the angle between the ray and the perpendicular to the edge of the core, when it reaches it;

- i_2: the same angle, once inside the core.

The formulas. Calculator in degrees (DEG).

- $\sin i_1 = (9/5) \cdot \sin i_0$

- $\sin i_2 = 0.8 \cdot \sin i_1$ (Snell's law, from 10 to 8 km/s)

- If the ray does not reach the core, it comes out at a distance $\Delta = 180° - 2 \cdot i_0$ from the focus.

- If it crosses the core, it comes out at $\Delta = 2 \cdot (i_1 - i_0) + (180° - 2 \cdot i_2)$: two stretches of mantle and one of core, seen from the centre of the Earth.

- If Δ comes out greater than 180°, the ray has passed to the other side: its distance is $360° - \Delta$.

1. The drawing and the ray that grazes the core

With the compass, draw the Earth with a radius of 9 cm and, with the same centre, the core, with 5 cm. Mark the focus at the top.

1. The ray that grazes the core satisfies $\sin i_0 = 5/9$. What is i_0?

2. Draw it with the protractor and the ruler. How many degrees from the focus does it come out? Measure it from the centre of the Earth and check it with the formula.

2. The rays that cross the core

One example worked in full and another half done. Keep four decimal places in the sines and two in the angles, and round only Δ.

i_0	sin i_1	i_1	sin i_2	i_2	mantle: $i_1 - i_0$	core: $180° - 2 \cdot i_2$	Δ
30°	0.9000	64.16°	0.7200	46.05°	34.16°	87.89°	156.2°
20°	0.6156	38.00°			18.00°		

Your group works out the rows it is given. Then compare each result with Table I in Lehmann's paper.

3. The empty band

Mark your arrivals on the large circle on the board.

1. Between what distances does no ray land?

2. Where do the arrivals pile up? How is that like a magnifying glass?

4. The hypothesis: a faster inner core

Add an inner core of 0.2205 radii (2 cm in your drawing) where the wave travels at 8.6 km/s. Two more angles: i_3, the ray's angle when it reaches its edge, and i_4, inside it.

- $\sin i_3 = 2.52 \cdot \sin i_2$

- $\sin i_4 = (8.6/8) \cdot \sin i_3$

- If the ray crosses the inner core, it comes out at $\Delta = 2 \cdot (i_1 - i_0) + 2 \cdot (i_3 - i_2) + (180° - 2 \cdot i_4)$.

- If sin i_4 comes out greater than 1, the ray cannot get in: it bounces off entirely, and comes out at $\Delta = 2 \cdot (i_1 - i_0) + 2 \cdot (i_3 - i_2)$.

i_0	i_1	i_2	i_3	i_4	Δ
14°					
13°					
15°					

1. Where do those rays land? Does the inner core explain the waves that arrived inside the shadow?

2. What happens with $i_0 = 15°$? Is it a calculator error?

5. Conclude

1. In the real Earth, without an inner core, the shadow would run from about 98° to about 145°. Why is the shadow in this model so far off? Think about how the rays travel in the mantle.

2. What is such a simple model for, if it does not get all the numbers right?

Practice

1. In the second class you worked out that the core holds one third of the Earth's mass. Explain why that does not contradict the P wave travelling more slowly in the core than in the mantle.

2. A wave passes from a rock where it travels at 6 km/s to another where it travels at 8 km/s. Say whether it moves closer to the perpendicular or away from it, and what would happen the other way round.

3. Put these discoveries in chronological order and say what evidence each one used: the inner core, the Moho, the liquid core and the depth of the core.

Try it yourself

If you want to, with a plain, clear glass full of water, away from the computer:

1. Put a spoon in it, resting on the rim and tilted to one side, not towards you.

2. Put your eyes level with the water and look at the spoon through the glass. What does it look like?

3. Turn the glass slowly. What happens to the break?

4. What does the spoon show well about the waves that cross the core? What can't it show?

If you don't have a plain glass (it doesn't work with a cut-glass or coloured one), look at the photo in the video.

For keen makers: a shadow like the core's

If you feel like trying it, carefully:

1. Find a room you can make dark. Before switching off the light, clear the floor; afterwards, move calmly.

2. Fill a plain, clear glass almost to the top, and put it less than a hand's span from a light-coloured wall (or from an open book standing upright).

3. Put your phone's torch four fingers' width from the glass, level with the water, pointing at the centre. Don't let the water touch the phone; if any spills, dry it before carrying on.

4. Look at the wall: on both sides of the glass, can you see two dark bands? Look over the top of the glass: can you see a very bright line in the middle?

5. Which part of the pattern is the shadow? And the magnifying glass?

If you can't make the room dark or don't have a plain glass, the simulation in the video shows what you would see.

Keep thinking

Others had already seen those waves inside the shadow, and had looked for another explanation for them. Today, records from thousands of seismometers are public: anyone can download them. What do you think they still hide?

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